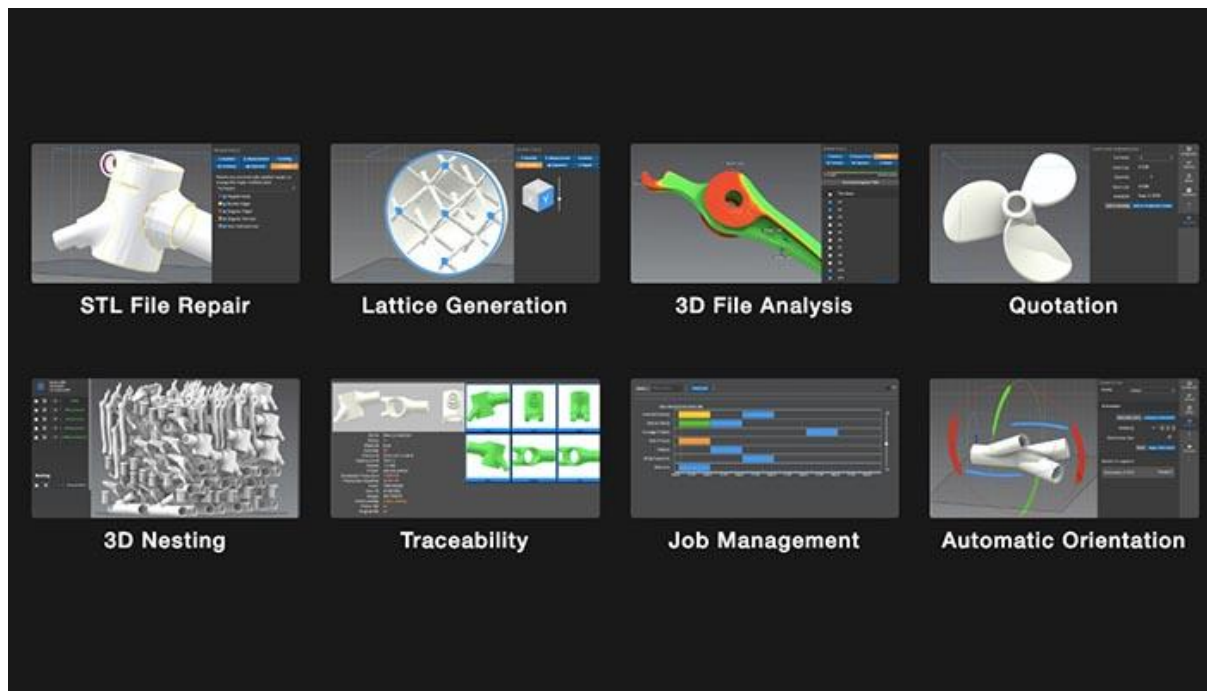




COURSE CONTENT

- 4.1.1 STL files**
- 4.1.2 Overview of RP Software's**
- 4.1.3 Other Popular Software's**

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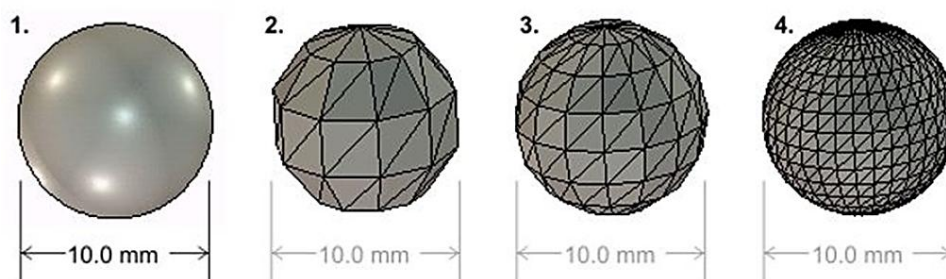
4.1.1 STL FORMAT

STL (Standard Tessellation Language) is a file format native to the STereo Lithography CAD software created by 3D Systems.

- Tessellation is piecewise approximation of surfaces of 3D CAD model using series of triangles.
- The STL file connects the surface of the model in an array of triangles and consists of the X, Y and Z coordinates of the three vertices of each surface triangle, as well as an index that describes the orientation of the surface normal.
- It is widely used for Rapid Prototyping (RP) and Computer-Aided Manufacturing (CAM). This file format is supported by many other software packages.
- STL files describe only the surface geometry of a three-dimensional object without any representation of colour, texture or other common CAD model attributes.
- The STL format specifies both ASCII and binary representations.
- STL files determines the resolution of the print. A model with a high resolution will have a smooth surface with a high level of detail but will take more time to be sliced.
- This format is used as industry standard in Rapid Prototyping.

<pre> solid print facet normal 0.00000e+00 1.00000e+00 0.00000e+00 outer loop vertex 0.00000e+00 0.00000e+00 2.00000e+01 vertex 0.00000e+00 0.00000e+00 0.00000e+00 vertex 1.00000e+01 0.00000e+00 2.00000e+01 endloop endfacet facet normal 0.00000e+00 1.00000e+00 0.00000e+00 outer loop vertex 1.00000e+01 0.00000e+00 2.00000e+01 vertex 0.00000e+00 0.00000e+00 0.00000e+00 vertex 1.00000e+01 0.00000e+00 0.00000e+00 endloop endfacet </pre>	
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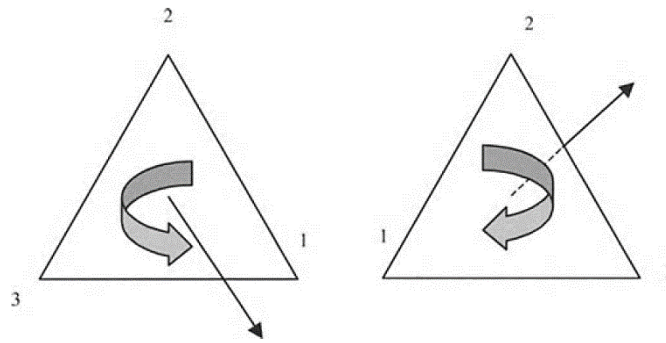
Sample STL file



STL file with different resolutions

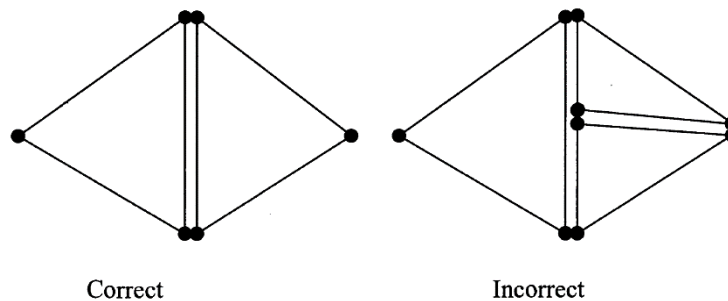
- Two very important requirements must be followed during STL file generation:
 - Data about triangle vertices must be stored in the file in an ordered fashion to identify interior or exterior surfaces. A clockwise vertex ordering defines an

interior surface, and an anti-clockwise ordering the exterior surface. The righthand rule is applied.



Righthand rule

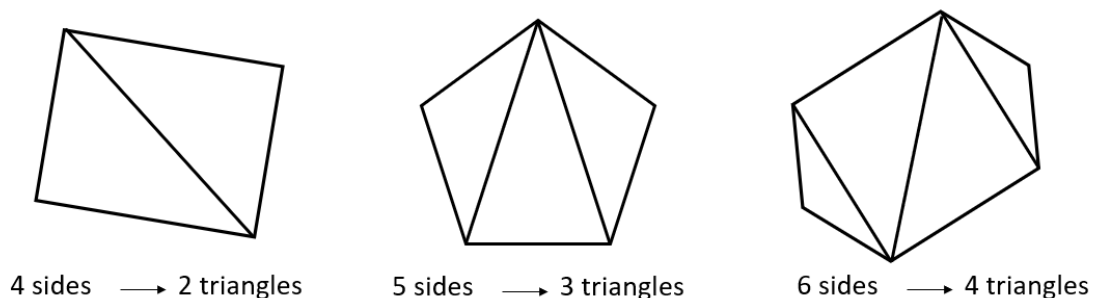
- A triangle must share exactly two common vertices with each adjacent triangle. This is known as the vertex-to-vertex rule.



Vertex-to-vertex rule

PROCESS OF APPROXIMATING SURFACES WITH TRIANGLES:

- Each 3D form is made out of polygons.
- A polygon is defined as a flat shape which is bounded by a closed circuit.
- Each polygon with n sides can be represented using $(n - 2)$ triangles



Why is it used in RP?

- The STL file format has become the standard data transmission format, and is the format required to interact with Quickparts.
- Slicing a part using STL format is easier compared to other formats, such as Boundary representation (B-rep) and Constructive Solid Geometry (CSG), which will need complex geometric reasoning and data conversion.
- Almost all of today's CAD systems are capable of producing an STL file.
- For the user the process is often as simple as selecting File, Save As, and STL.

- In all cases, export your STL file as a Binary file. This saves on time and file size.
- There is absolutely no loss of data whatsoever due to compression and encryption involved in the RP file format.
- If you create a RP file from a STL file and then convert it back to a STL file, you will end up with exactly the same STL file.

4.1.2 OVERVIEW OF RP SOFTWARE'S

The modern engineering ecosystem relies heavily on digital tools that allow designers, engineers, and medical professionals to visualize, modify, analyze and prepare 3D data. Three important tools used widely in engineering and biomedical applications are:

- SolidView – For viewing, measuring, and annotating CAD models
- Materialise Magics – For preparing and repairing 3D files for additive manufacturing
- Materialise Mimics – For converting CT/MRI scans into accurate 3D anatomical models

These tools bridge the gap between design, manufacturing, healthcare, and research.

1. SOLIDVIEW

SolidView (now called Freeform) is a 3D visualization and communication software that allows users especially those without CAD expertise to view, inspect, and interact with CAD models. It removes the dependency on complex CAD software for everyday engineering communication.

Key Features:

a) 3D Model Viewing:

- SolidView can open various CAD file formats such as STL, IGES, STEP, and DXF, enabling compatibility with multiple CAD systems.
- It provides smooth real-time rotation, zooming, and panning, allowing easy examination of complex models.
- The software uses lightweight rendering, making it capable of loading and displaying large assemblies efficiently.

b) Measurement & Inspection:

- The software supports essential measurements such as linear distances, angles, radii, and diameters.
- Users can create section views to analyze internal features of a model.
- Advanced versions allow volume and surface area calculations for quick engineering checks.

c) Markup & Collaboration:

- SolidView allows users to add annotations, labels, comments, and simple dimensions directly on the model.
- These marked views can be exported as images or included in reports for sharing with team members or vendors.
- It is a useful tool during design reviews and communication with stakeholders who do not use CAD software.

d) Model Translation:

- The software can convert CAD files into neutral or lightweight formats for easier sharing.

- This helps protect the original CAD files by distributing simplified versions for viewing only.
- It supports collaboration across teams using different software tools.

Industrial Use Cases

- Manufacturing engineers inspecting designs before fabrication
- Managers reviewing models without CAD training
- Quality inspection teams checking tolerances
- Communication between OEMs and vendors
- Technical illustrators preparing manuals and part catalogs

2. MATERIALISE MAGICS

Materialise Magics is a comprehensive AM data preparation software, considered the industry standard in 3D printing workflow. It provides powerful features to repair, modify, optimize, and prepare digital models before they are sent to a 3D printer.

Key Features:

a) CAD Import & Analysis:

- Magics supports importing a wide range of CAD formats and converts them into mesh data for editing.
- It provides tools to analyze model quality, thickness, and errors before printing.

b) STL Repair & Optimization:

- The software can repair common mesh issues such as holes, bad edges, inverted normals, and intersecting triangles.
- Magics ensures the model becomes watertight and printable.
- Optimization tools help improve print quality by refining the mesh where needed.

c) Build Preparation for 3D Printing:

- Users can orient the model to achieve better strength, surface finish, and reduced support usage.
- Automatic and manual support generation tools help prepare models for metal and polymer 3D printing.
- Nesting tools allow arranging multiple parts efficiently on the build plate, reducing time and material usage.

d) Manufacturing Workflow Tools:

- Magics provides features for production planning, part labeling, job documentation, and quality inspection.
- Lattice and lightweighting tools help reduce weight while maintaining structural integrity.
- It supports integration with various industrial 3D printing systems.

Industrial Applications

- Aerospace and automotive printing
- Medical implants & prosthetics
- Tooling and jigs
- Industrial prototypes
- High-volume AM service bureaus

3. MATERIALISE MIMICS

Materialise Mimics is specialized software used for converting medical imaging data (CT, MRI, micro-CT) into accurate 3D anatomical models. It is widely used in healthcare, biomedical engineering, surgery planning, and research.

Key Features:

a) Medical Image Processing:

- Mimics imports DICOM data from CT, MRI, and other imaging systems.
- It provides tools for segmentation to isolate bones, organs, blood vessels, soft tissues, or implants.
- Users can visualize anatomical structures in 2D slices and 3D models.

b) 3D Model Generation:

- The software converts segmented regions into accurate 3D surface models representing patient-specific anatomy.
- The models can be smoothed, refined, and edited as needed.
- Mimics ensures high accuracy suitable for engineering analysis and medical use.

c) Measurements & Analysis:

- Users can measure distances, angles, thickness, volume, and shape directly on the anatomical model.
- These tools assist in diagnosis, pre-operative planning, and research.

d) Export & Integration:

- Models can be exported in STL format for 3D printing or simulation.
- Mimics integrates with Materialise 3-matic for advanced mesh optimization suitable for FEA or CFD.
- The models can also be exported to other engineering tools for implant design or biomechanical studies.

Applications:

- Preoperative surgical planning
- Designing patient-specific implants and prosthetics
- Medical education and training
- Research in biomechanics
- Production of anatomical models for teaching and diagnosis

4. MAGICS COMMUNICATOR:

Magics Communicator is a lightweight 3D viewer designed to help users view, inspect, and discuss 3D files that have been prepared using Materialise Magics. It allows engineers, clients, and production teams to review models without needing the full Magics software, making it ideal for communication and approval workflows in AM.

Key Features:

a) Viewing STL and CAD Files:

- Magics Communicator can open STL and various CAD files that have already been processed or exported from Magics.

- Users can visually inspect the geometry, check mesh quality, and explore different viewing modes to understand the design clearly.
- The tool is optimized to load files quickly so even large models can be examined without performance issues.

b) Review of Part Orientation, Supports, and Build Strategies:

- The software allows users to see how the part has been oriented for 3D printing in Magics.
- It also displays all support structures, enabling reviewers to verify whether the part is properly supported for successful printing.
- Users can examine build strategies such as layer direction and the positioning on the build platform, helping ensure print reliability.

c) Annotation and Comments:

- Magics Communicator includes simple markup tools that allow users to add comments, highlight areas, and point out issues directly on the model.
- These annotations help in clearly communicating necessary corrections or approvals between engineers and clients.
- The comments can be saved and shared so everyone involved can follow the same feedback trail.

d) Lightweight Collaboration Tool:

- Because Magics Communicator does not require a full Magics license, it is very accessible for external partners, customers, and non-technical stakeholders.
- It ensures that all reviewers see the same 3D preparation details support structures, orientations, and build settings exactly as prepared by the AM engineer.
- This reduces errors and misunderstandings during the manufacturing process.

4.1.3 OTHER POPULAR SOFTWARE'S

Following are some Popular Software Used in 3D Modeling, Additive Manufacturing

3D Modelling Softwares

- SolidWorks
- CATIA
- Siemens NX
- PTC Creo
- Autodesk Inventor
- Rhino 3D
- Blender
- FreeCAD

Slicing & Printer Preparation Software

- Ultimaker Cura (offline mode)
- PrusaSlicer
- Lychee Slicer
- Simplify3D
- FlashPrint

Internet-Based AM Softwares

- Materialise CO-AM Platform
- Autodesk Fusion 360
- Onshape
- 3DPrinterOS
- GrabCAD Print
- Ultimaker Digital Factory
- EOSCONNECT
- HP 3D Center
- Formlabs Dashboard
- nTopology
- Siemens Teamcenter (Cloud AM workflows)
- AMFG (Additive MES)
- MakerBot CloudPrint
- Shapr3D Cloud Sync
- Tinkercad